Who Decides Washington?

Who Returns the Ballot? Age Composition in Washington's Odd-Year Electorate, 2021–2025

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AI-assisted analysis; every figure is independently reproducible from public records via the cited open-source scripts (e.g. `scripts/verify_who_decides_wa.py`). Contact: kirby@tikorconsulting.com.

Abstract

Many of Washington's local offices are filled in odd-numbered November elections, when turnout is far below presidential-year turnout. This paper asks *who returns those ballots*. Joining Washington's statewide voter-registration database — a 5.51-million-voter roll linked to 27.1 million individual vote-history records and each voter's year of birth — it measures the age composition of every November general electorate from 2021 through 2025. The central finding is descriptive: Washington's odd-year ballot-return electorate is markedly older than its presidential electorate. Voters 65 and older were 36.7%, 40.2%, and 40.3% of the 2021, 2023, and 2025 odd-year electorates, versus 28.5% in the 2024 presidential electorate; voters 18–29 fell from 14.2% in 2024 to roughly 7.6% off-cycle. The result survives validation against certified ballot counts, a formal worst-case bound for voters missing from the current-roll reconstruction (confirmed against a second, closer-in-time roll snapshot), alternative birth-year imputations, all 39 counties, and the exclusion of any single off-year; the off-year electorate is older than the registered roll and the citizen voting-age population as well. Individual records further show it is largely the presidential electorate's *habitual core* — 92–97% of off-year voters also vote in presidential years, while the peripheral voters who fall away off-cycle are disproportionately young. The paper measures ballot return, not votes cast in specific down-ballot contests, and does not estimate partisan or policy consequences. Its contribution is a validated, individual-record measurement of age composition across the presidential–midterm–off-year salience gradient in a universal vote-by-mail state where formal ballot-access friction is comparatively low.

Keywords: election timing; voter turnout; local elections; age representation; voter files; Washington; vote by mail; off-cycle elections.

The question

Most coverage of "turnout" asks how *many* people vote. The more consequential question for how a state is governed is *who* — because the answer changes with which election you look at. Washington runs many of its most local offices — city councils, school boards, port and fire commissions, and many county and judicial races — in **odd-numbered, off-cycle Novembers** (RCW 29A.04.321 limits the odd-year statewide general election chiefly to city, town, and district offices, certain county positions, and state measures), when only **~38%** of registered voters return a ballot. This paper measures, at the individual level, who that ~38% is.

The short answer: **the observable odd-year ballot-return electorate is markedly older than the presidential one.** It is not a smaller copy of the presidential electorate; it is an older one.

What is measured. This paper measures **ballot return** — whether a registered voter cast a November ballot — at the individual level. It does *not* observe whether that voter marked a specific city-council, school-board, port, fire, judicial, or county contest; voter-file records show participation in an *election*, not a *contest* (Lucero et al. 2025). That distinction matters most for the **even-year counterfactual**: if local races were consolidated onto longer even-year ballots, some added voters would return a ballot but skip the local race. For the **odd-year electorate described here**, ballot return is a closer proxy for local-election participation, because Washington's odd-year general election is dominated by local and district contests, with only occasional statewide measures — but it remains an election-level measure, not a contest-level one. Appendix F treats roll-off as the live counterargument.

Data and validation

Source. `voting_history` (27.1M records) joined to `voters` (year of birth, ~100% coverage) from Washington's standard statewide VRDB extract (April 2026; provenance and use terms in Appendix B). Age is taken as election year − birth year; cohorts are assigned per election. November generals only, 2021–2025.

A hostile reading of any voter-file study is: *you did not measure the electorate; you measured the electorate that remains observable in your current voter file.* Because the vote-history table is keyed to a current (2026) roll, a voter who cast a ballot in 2021–2024 but has since died, moved, or been canceled is absent. If those departed voters were a random slice, coverage loss would be harmless; if they skew young, the finding would be inflated. Both are checkable, and the answers favor the finding.

Coverage is high, and the lowest-coverage case is bounded directly. The reconstruction is benchmarked against certified WA Secretary of State ballot counts (`results.vote.wa.gov`). "In file" is distinct voters in the cumulative vote-history table for that election; "Analyzable" additionally requires a match to the April 2026 roll with a year of birth. Analyzable coverage ranges from

90.8% in 2021 to 99.6% in 2025 and improves sharply over time; the lowest-coverage election is the 2021 off-year, which is why the analysis reports an explicit worst-case bound (below) rather than assuming the missing voters resemble observed voters. The validation target is *certified ballots counted*; the vote-history file is a voter-level reconstruction of ballot credit, expected to track the certified count closely but not to be mechanically identical to it.

Election	Type	Official ballots counted	In file	Analyzable (roll + YOB)	Analyzable / official
Nov 2021	Off-year	1,896,481	1,828,231 (96.4%)	1,722,262	**90.8%**
Nov 2022	Midterm	3,067,686	3,025,352 (98.6%)	2,884,966	**94.0%**
Nov 2023	Off-year	1,758,084	1,731,431 (98.5%)	1,686,656	**95.9%**
Nov 2024	Presidential	3,961,569	3,958,965 (99.9%)	3,880,070	**97.9%**
Nov 2025	Off-year	2,001,425	1,995,509 (99.7%)	1,993,505	**99.6%**

The observable attrition component skews old, and the full residual is bounded. A second roll snapshot (`voters_20230901` , 5.29M rows, Sept 2023) partially characterizes voters who cast a past ballot but are absent from the April 2026 roll. Among these **observable attrition cases**, seniors are substantially overrepresented:

Missing-from-current-roll voters	65+ share	18–29 share
Cast a ballot Nov 2021 (n≈106K)	**60.4%**	6.8%
Cast a ballot Nov 2022 (n≈140K)	**60.3%**	7.1%
Cast a ballot Nov 2023 (n≈45K)	**69.0%**	4.5%

Roll departure is age-loaded more broadly (of the 504K voters, 9.5%, who left the roll 2023→2026, 33.1% were 65+ vs 23.9% of those who stayed — consistent with mortality and age-correlated mobility/cancellation). This evidence suggests the current-roll reconstruction **likely understates** the senior share of past electorates. But because not every missing ballot can be characterized from the second snapshot, a **formal bound** is also reported that assumes nothing about the residual: the 65+ share of the *full certified electorate* if every unobserved residual voter (official − analyzable) were under 65, and if every one were 65+.

Off-year 65+ share of the certified electorate	all missing < 65 (min)	observed	all missing 65+ (max)
Nov 2021 (residual 9.2%)	**33.4%**	36.8%	42.6%
Nov 2023 (residual 4.1%)	**38.6%**	40.2%	42.6%
Nov 2025 (residual 0.4%)	**40.2%**	40.3%	40.6%

"Observed" is the 65+ share of the analyzable electorate from the validation table (2021 = 36.8%; the composition table's 36.7% adds a "registered on/before the election" filter — a 0.1-point difference). Min/max apply the extreme assumption to the residual = official − analyzable.

Even under the minimum assumption — every unobserved off-year ballot cast by someone under 65 — each off-year electorate (**33.4% / 38.6% / 40.2%** 65+) remains older than the presidential electorate under *its* maximum assumption (**≤30.0%**, i.e. even if every missing 2024 ballot were 65+). The substantive result therefore does not depend on how the residual missing voters are allocated. Two claims, kept distinct: the observed composition estimates are **likely conservative** given the attrition evidence, and the **formal worst-case bound** preserves the finding regardless.

A direct check: composition barely moves under a closer-in-time roll. The bound above is agnostic about the missing residual; a complementary check asks whether reconstructing from a *current* roll distorts composition at all. The database's Sept-2023 snapshot (`voters_20230901`) still contains voters who have since left the 2026 roll, so it reconstructs the 2021–2023 electorates from a roll much closer in time. The two reconstructions agree to within ~1.4 points — and what movement there is runs *upward*, consistent with the current roll modestly under-counting seniors:

Election	Current-roll 65+	Sept-2023-snapshot 65+	Δ
Nov 2021 (off-year)	36.8%	38.1%	+1.4
Nov 2022 (midterm)	31.0%	32.4%	+1.4
Nov 2023 (off-year)	40.2%	41.1%	+0.9

The snapshot — which recovers older voters the current roll has since dropped — sits slightly *higher*, not lower, so the composition finding is not an artifact of current-roll reconstruction. The 2021 and 2022 comparisons are the cleanest checks because the September-2023 snapshot postdates those elections. The 2023 comparison should be read more cautiously: the snapshot predates the November-2023 election by two months and so misses late-2023 registrants who voted that November; if those late registrants skew younger, as new registrants often do, the September-snapshot 2023 estimate may be biased slightly *upward*. Even so, the check is reassuring — the closer-in-time reconstruction does not make any electorate younger than the current-roll one, and the 2021–2022 comparisons point the same way.

What the data shows

The robust cut is **composition** — each age group's share of the ballot-returning electorate. It needs no turnout denominator, rests on three consistent off-year cycles, and (per the validation above) survives an explicit worst-case bound.

Election	Type	18–29	30–44	45–64	65+
Nov 2024	Presidential	14.2%	24.9%	32.4%	28.5%
Nov 2022	Midterm	10.4%	22.9%	35.7%	31.0%
Nov 2021	Off-year	7.5%	19.7%	36.0%	36.7%
Nov 2023	Off-year	7.4%	19.2%	33.2%	40.2%
Nov 2025	Off-year	8.0%	19.9%	31.7%	40.3%

- **The off-year electorate is a senior-plurality electorate, stably so.** Voters 65+ are **~37–40%** of it (36.7 / 40.2 / 40.3% across 2021 / 2023 / 2025) versus **28.5%** presidential; the 18–29 share falls from **14.2%** to **~7.6%**.
- **The senior-to-youth ratio roughly triples off-cycle** — from about **2:1** (presidential) to **~5:1** (off-year), with the midterm between (31.0% 65+). The *off-year* result is stable across three cycles; the presidential and midterm points are single elections (2024, 2022), so the ordering is consistent with a salience gradient but should not be read as a smoothly estimated curve.

Residents, registrants, voters: an escalating age ladder. The off-year electorate is older than the presidential one, older than the registered roll, older than the citizen voting-age population, and older than all adult residents. Setting the ballot-returning electorates beside the roll (April 2026), the citizen voting-age population, and all adult residents (both ACS 2020–24) gives a monotonic ladder:

Population	18–29	30–44	45–64	65+	Median age
WA adult residents (ACS 2020–24)	20.0%	28.3%	30.5%	21.1%	~46†
WA citizen voting-age population (ACS 2020–24)	19.8%	26.7%	30.9%	22.6%	~47†
Registered roll (April 2026)	16.7%	27.2%	29.8%	26.3%	48
2024 presidential ballot-returners	14.2%	24.9%	32.4%	28.5%	52
Off-year ballot-returners (2021/23/25 avg)	7.6%	19.6%	33.6%	39.1%	59

† Age composition from ACS 2020–24 five-year (the most recent five-year vintage; the 2024 one-year is consistent), reproduced by `scripts/acs_wa_adult_age.py`: total 18+ residents from table B01001, and the **citizen voting-age population (CVAP)** — the eligible-electorate benchmark, which excludes non-citizens — from table B29001. Median ages for the two ACS rows are interpolated from the age brackets; the roll and ballot-returner medians are computed from year of birth and are integer-year (±1) approximations. Roll and ballot-returner figures from the VRDB (`scripts/verify_who_decides_wa.py` #23).

The 65+ share climbs **21.1% → 22.6% → 26.3% → 28.5% → 39.1%** across the five; the 18–29 share falls **20.0% → 19.8% → 16.7% → 14.2% → 7.6%**. The median off-year ballot-returner is **59 —** about a decade older than the median registered voter (48) and over a decade older than the median citizen-voting-age adult (~47). Residents, eligible citizens, registrants, and voters are distinct populations, and it is participation that moves the composition: the registered roll's senior share is stable and low (26.3% on the full April 2026 roll; ~22–25% on the per-election eligible roll in each of these years) while the off-year returner share reaches ~40%.

One number for "how unrepresentative." Collapsing the ladder into an index of dissimilarity between each electorate's age distribution and the citizen voting-age population (half the summed absolute cohort differences; 0 = identical) turns the gradient into a single figure: **7.4** for the 2024 presidential electorate, **13.2** at the midterm, and **18.5–19.9** across the three off-years **—** the off-year electorate is roughly **2.5× as age-unrepresentative** of the eligible population as the presidential one. The index is cohort-based, so its level depends on the age bins used; its value here is comparative, not absolute.

Recorded gender shows a much smaller secondary pattern: the electorate is majority recorded-female throughout, rising from 52.5% in the 2024 presidential electorate to 53.0–53.1% in the off-year electorates, concentrated among older voters. Because the shift is small and administrative gender is not the paper's focus, the analysis treats age as the primary dimension.

The mechanism is differential participation. The within-cohort rates below make it concrete — 18–29 participation falls from **58.4%** (2024) to an off-year **~16%** while 65+ falls only from **88.3%** to **~61%** — but they *corroborate* the composition finding, they do not carry it:

Within-cohort participation rate — current-roll reconstruction, not official turnout:

Election	Type	18–29	30–44	45–64	65+	All
Nov 2024	Presidential	58.4%	68.6%	80.1%	88.3%	75.0%
Nov 2022	Midterm	36.4%	52.5%	69.6%	84.6%	62.4%
Nov 2021	Off-year	16.5%	28.6%	43.1%	65.6%	39.4%
Nov 2023	Off-year	14.5%	24.6%	37.1%	59.0%	34.9%
Nov 2025	Off-year	16.4%	27.0%	39.0%	59.3%	36.7%

These are not official turnout rates. The denominator is the age-eligible **April 2026 roll**, not the election-day registered cohort, so a later (larger) roll mechanically depresses them — the "All" column (e.g., 75.0% in 2024) runs below Washington's official general-election turnout (39.38% 2021, 63.82% 2022, 36.41% 2023, 78.95% 2024, 39.24% 2025). Read the table as a current-roll *reconstruction* of within-cohort participation. It also rests on a single presidential (2024) and single midterm (2022) cycle.

Sensitivity

The finding does not depend on the cohort boundaries, the birth-year imputation, any single off-year, or King County.

Finer cohorts. Splitting the endpoints sharpens the pattern: the 75+ share rises from **11.8%** (presidential) to **16.8–18.3%** off-year; the 18–24 share falls from **7.7%** to **~3.7–4.0%**.

Election	Type	18–24	25–29	30–44	45–64	65–74	75+
Nov 2024	Presidential	7.7%	6.5%	24.9%	32.4%	16.7%	11.8%
Nov 2022	Midterm	5.3%	5.1%	22.9%	35.7%	19.4%	11.7%
Nov 2021	Off-year	3.7%	3.8%	19.7%	36.0%	23.3%	13.4%
Nov 2023	Off-year	3.7%	3.6%	19.2%	33.2%	23.4%	16.8%
Nov 2025	Off-year	4.0%	4.0%	19.9%	31.7%	22.1%	18.3%

Birth-year imputation. The file supplies year of birth, not full date, so the main analysis uses age = election year − birth year — equivalent to assuming the birthday has occurred by the November election. This may classify voters with late-November or December birthdays as one year older than their exact age on Election Day. As a sensitivity test the 65+ share is recomputed under the opposite extreme — every voter treated as if their birthday had *not* yet occurred (a Dec-31 imputation) — which moves the off-year 65+ share by **≤2.4 points** (e.g., 2021: 36.8% → 34.3%; 2025: 40.3% → 38.2%) and leaves the presidential/off-year gap intact (the presidential share moves too, 28.5% → 26.7%). Because November falls late in the calendar year, the true value should lie *closer to the main convention* than to this all-younger extreme, absent unusual late-year birthday clustering.

Off-year stability, and statewide measures. The three off-years bracket the same result (65+ share 36.7 / 40.2 / 40.3%) despite *different* statewide ballot content: 2021 and 2023 carried only the state's non-binding tax "advisory votes" (since repealed), and 2025's only statewide item was a single fiscal constitutional amendment (SJR 8201, on investing the WA Cares trust fund — the only statewide measure on the 2025 ballot, per the WA Secretary of State). None is a high-salience mobilizing contest, and the stability of the 65+ share across all three is the direct evidence that none drives the composition. Excluding any one off-year leaves the conclusion intact.

Geography. King County (largest, youngest) runs younger than the rest at every salience level, but the gradient is present everywhere and steeper outside the urban core.

65+ share of electorate	2024 Pres	2021 Off	2023 Off	2025 Off

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King County	23.0%	28.7%	32.2%	30.7%
Rest of state	30.7%	40.4%	43.5%	45.0%
Metro counties[1]	26.4%	34.3%	37.5%	37.3%
Rural counties	37.7%	46.7%	51.5%	54.2%

[1] Metro = the ten most-populous counties (King, Pierce, Snohomish, Clark, Spokane, Thurston, Kitsap, Whatcom, Benton, Yakima); rural = the remaining 29.

The presidential→off-year shift toward seniors is positive in **all 39 counties** (full breakdown in Appendix E).

Interpretation: mechanism, lever, and policy caution

It is behavior, not the rolls. A symmetric two-factor decomposition (Kitagawa–Das Gupta; `scripts/diag_turnout_decomposition.py`) of the 2024→2025 change in the 65+ share **supports the behavioral interpretation**: of the **+11.8-point** rise, **+10.9 (92%) is the turnout-rate effect** and only **+0.9 is roll composition**; for 18–29 the split is **−6.0 of −6.2 (97%) behavioral.** The split is stable across all three off-years — the turnout-rate (behavior) effect accounts for **92% (2025), 95% (2023), and 79% (2021)** of the 65+ rise, with the roll-composition effect small and, in 2021 and 2023, slightly *negative* (a marginally younger roll worked against the shift, so behavior more than accounts for it). Both endpoints of each pair are drawn from nearly the same near-current roll, so the current-roll denominator bias is common to both and largely cancels in the difference; and the older-skewing attrition above would, if corrected, *raise* the senior rate, not shift weight to the roll term. The skew is a turnout/salience phenomenon — which is why the lever is **election timing**, not registration policy.

The off-year electorate is the presidential electorate's habitual core. Among **analyzable current-roll voters**, individual records make surge-and-decline concrete in a way aggregate data cannot. Roughly **92–97%** of each off-year electorate also cast a 2024 presidential ballot, while only **42–48%** of 2024 presidential voters turned out in a given off-year — the off-year electorate is close to a *recurring core* of the presidential one. Splitting 2024 presidential voters by whether they *also* voted the 2023 off-year shows who stays and who leaves: the **habitual core** (voted both; 1.6M) is **42.8% 65+ and 6.1% under 30**, while the **presidential-only drop-off** (2.2M) is **18.0% 65+ and 20.2% under 30**. The voters who appear mainly when a presidential race is on the ballot are disproportionately young; the off-year electorate is what remains when that peripheral electorate falls away. Off-year voters also have **older current registration records** — a median of roughly **16–17 years** since registration date versus **12** for the presidential electorate — though registration-record age is not a perfect measure of lifetime civic attachment. Because this overlap is computed among analyzable current-roll voters, it should be read alongside the survivorship checks above: attrition removes some past voters from the panel, especially older voters who died or moved before the 2024 comparison point.

The lever. The quasi-experimental evidence that on-cycle timing reshapes the electorate is strong. Anzia (2014) and Hajnal & Trounstine (2005): off-cycle elections shrink the electorate and make it less representative, shifting outcomes toward organized, high-propensity groups. Hajnal, Kogan & Markarian (2022), using individual micro-targeting data, find California's move to on-cycle municipal elections roughly **doubles** local turnout and makes the electorate considerably more representative by age, race, and partisanship. Lucero et al. (2025), in a switcher-city survey, report voters over 45 at **58.4%** of the off-cycle vs **49.7%** of the presidential- concurrent electorate (and, citing Hajnal et al., a ~**22-point** over-55 gap). This paper measures Washington's *gap* — ~40% off-year 65+ vs ~28.5% presidential — but not what consolidation would do here: **if Washington followed the pattern those studies document, the expected effect of moving local races onto even-year Novembers would be a substantially larger and younger local electorate** — a plausible extrapolation from California city-election evidence, not a Washington simulation.

Policy caution. The mechanism literature gives reason to *expect* that a smaller, older, more-organized off-cycle electorate produces different policy (Anzia 2014, on public-employee compensation; Kogan, Lavertu & Peskowitz 2018, on school-district spending). But the evidence is not settled: **Ornstein (2024)**, using California's 2018 on-cycle mandate (SB 415) across 236 local governments, finds the expected turnout and diversity gains but **no** detectable downstream effect on descriptive representation, the candidate pool, the incumbency advantage, housing policy, or public-employee salaries. **This paper does not adjudicate that debate.** It estimates participation and composition; the composition result is not evidence of policy capture. (The full taxonomy of objections — preference-intensity, ballot dilution / roll-off, and age-as-proxy — is in Appendix A.)

What this paper does not claim, and limits

- **No partisan consequence.** Washington does not publish party of record, so this paper stops at participation and composition. The party-registration companions (New York, Idaho) test that.
- **No policy-outcome claim.** The composition finding is not evidence of policy capture, union influence, or changed local

outcomes; the literature is mixed (above) and it is not tested here.

- **No contest-level claim.** This paper measures ballot return in November, not participation in a specific local contest (Lucero et al. 2025). Appendix F estimates even-year *contest*-level roll-off as a first look at the sequel.
- **No individual-level causal claim** about why any person votes or abstains.
- **Age vs cohort.** Five cycles (2021–2025) cannot separate a *life-cycle* effect (older people vote off-cycle) from a *cohort* effect (a durable high-propensity generation that is currently old). The habitual-core result is consistent with either; distinguishing them needs a longer individual panel.
- **Cycle coverage.** The vote history begins in 2021, so the presidential row rests on **2024 alone** and the midterm on **2022 alone**; only the off-year row averages three cycles. King County 2020 presidential is not loaded and is excluded (all figures are 2021+).
- **Uncertainty.** The VRDB figures are near-population counts and carry no sampling error; the ACS resident/CVAP rows are 5-year *estimates* with margins of error (small at this geography and aggregation but nonzero), and the county roll-off correlation (n=39) is reported without a confidence interval — it is descriptive, not inferential.
- **Rates are a current-roll reconstruction**, not official turnout (above); the **share-of-electorate** figures, which need no denominator and are a bounded and validated estimate, carry the finding. Methods detail in Appendix C.

Appendices

Appendix A — The objections, in full

The most obvious objection is the weakest: **this is voluntary participation, not disenfranchisement.** Correct, and the right frame. Washington votes entirely by mail, prepaid, with automatic and same-day registration; no one is kept from the off-year ballot, and the remedy (moving local races onto even-year ballots) is an ordinary scheduling choice. This is a **design** question, not a rights question. But the benign reading explains *why* the off-year electorate is older; it does not make it less so, and the descriptive gap stands regardless of how one judges it. (It is also what makes Washington an informative case: with the usual access-cost explanation for youth drop-off weaker in a prepaid, same-day-registration state, the age gap is harder to attribute to friction.) Three stronger objections deserve direct treatment.

1 — The off-year electorate as a preference-intensity filter, not a defect. Perhaps the people who return low-salience local ballots are more informed, more locally attached, and more directly affected by property taxes, schools, and utility districts — and presidential-year voters, drawn by the national contest, may be *less* informed about local races and more likely to use national cues. This is the genuine normative crux. Three points bound it. (a) It is an argument about the *quality* of the marginal voter, which this participation study does not measure or resolve. (b) It cuts against a large literature finding on-cycle electorates *more* representative of the underlying population by age, race, and partisanship (Hajnal, Kogan & Markarian 2022) — closer to the community the government serves. (c) The information deficit it posits is itself partly an artifact of off-cycle timing (thin media coverage), so it is not clearly exogenous to the scheduling choice.

2 — Ballot dilution / down-ballot roll-off. If local races move to even years, more people receive the ballot but some skip the local contest, so the contest-level electorate grows by less than total turnout. This is measured directly for Washington (Appendix F): in the 2024 even-year general, roll-off (the share of counted ballots casting no vote in a contest) was **~3–7%** for partisan statewide offices and ballot measures, **~16–17%** for *contested* nonpartisan statewide contests (Supreme Court, Superintendent of Public Instruction), and **~34%** for *uncontested* ones — nonpartisan judicial races being the closest even-year analog to the local nonpartisan races (city council, school board) consolidation would add, and likely an upper bound (voters know least about judges). This exceeds the ~2–10% classic estimate (Wattenberg, McAllister & Salvanto 2000, who attribute roll-off substantially to *information*), and it is not assumed to be age-neutral for WA local contests, which remains untested. **Even so, the likely net direction is enlargement**, though its size depends on whether local races move onto a presidential- or a midterm-year ballot (Appendix F): even at the worst observed roll-off (34%), the even-year deciding electorate is ~52% of registered on a presidential ballot and ~42% on a midterm ballot, both above the ~38% who turn out off-cycle today. Enlargement survives substantial roll-off unless local roll-off runs far above the contested-nonpartisan benchmark.

3 — Age is not a clean proxy for whose interests matter. Seniors are heavily affected by local taxes, emergency services, transit, utilities, public safety, housing supply, and school levies, and no claim is made that younger people have more legitimate interests (an earlier draft's "most affected" phrasing overreached and is retracted). The narrower claim is **representational**, and stays entirely within the registered electorate the data measures: among registered voters, off-year ballot-returners are substantially older than presidential-year ballot-returners (median age ~59–60 vs 52, and ~a decade older than the median registrant, 48). If local offices are chosen by a markedly older subset of the registered electorate, the preferences counted at the ballot box differ systematically by age. The gap extends beyond registrants to the eligible and resident populations: the off-year electorate's 65+ share (~39%) is over **1.7 times** the **22.6%** of citizen voting-age Washingtonians who are 65+ (ACS 2020–24 CVAP) — and nearly double the 21.1% of all adult residents — while its 18–29 share (~7.6%) is under two-fifths of the ~20% share in both benchmarks (§ What the data shows).

Whether that is desirable, tolerable, or problematic is a normative question; the empirical contribution here is to measure its size.

Appendix B — Data access and privacy

- **What was obtained.** Washington's standard statewide **VRDB extract** — the single public extract the Secretary of State publishes (registrants plus cumulative vote history), regenerated monthly. This study uses the **April 2026 extract** (requested April 8, 2026). By statute the public file is limited to each voter's name, address, political jurisdiction, gender, **year of birth**, voting record, registration date, and registration number, and **no other information from voter-registration records is available for public inspection or copying** (RCW 29A.08.710) — the statutory basis for year-of-birth-not-full-date. Its use is restricted to elections and political purposes and **may not be used for commercial purposes** (RCW 29A.08.720); the file is **not redistributable**, with penalties at RCW 29A.08.740. Full provenance and access date: Data Sources & Reproducibility.
- **Year of birth, not date of birth.** Consistent with the statute, the extract supplies **year of birth only**; full date of birth is protected information that Washington withholds from the public voter database, and the legislature further strengthened voter-data protections in 2026 (SB 5892 / Ch. 213, Laws of 2026, eff. Mar. 25, 2026). In our file every birth value resolves to a July-1 sentinel, confirming year-only granularity — **no full date of birth was obtained, stored, or used.**
- **What is released.** Only aggregate cohort counts, with cell sizes in the thousands to millions; no individual-level records, addresses, or names. The analysis scripts emit aggregates only, and the repository's product layer additionally enforces a PII firewall (`src/wa_analyzer/product/firewall.py`). Only **citations and code, not data** are published — the only lawful option for the voter file.

Appendix C — Methods

- **Source and unit.** `voting_history` (27.1M records) joined to `voters` (year of birth, ~100% coverage), cohorts assigned per election. November generals only. The unit is **ballot return**, not participation in a specific contest.
- **Age convention.** With year of birth, age = election year − birth year (equivalently, birthday assumed reached by November). This is a ±1-year approximation of true age; the birth-year-imputation sensitivity (Sensitivity) shows the alternative extreme moves the off-year 65+ share by ≤2.4 points and leaves the gap intact.
- **Composition vs rates.** The within-cohort *participation rates* are a current-roll reconstruction (denominator = the April 2026 roll, not the election-day registered cohort), so they are not official turnout and are reported only to show the mechanism. The **share-of-electorate** figures, which need no denominator, carry the finding.
- **Coverage and bounding.** Coverage of the analyzable electorate against certified counts is 90.8–99.6%. Two distinct claims: the observable attrition component skews old, so the observed estimates are *likely conservative*; and, assuming nothing about the residual, a formal worst-case bound (every missing voter under 65) still keeps each off-year 65+ share above the presidential one, so the finding does not depend on the residual's composition. The Idaho companion reports **no** rates, because a ~13% roll contraction since 2024 inflates them past use — a difference in data, not method.
- **Decomposition.** The behavior-vs-rolls split is the symmetric two-factor (Kitagawa–Das Gupta) standardization (Appendix D).
- **Reproduction.** `scripts/verify_who_decides_wa.py` re-derives every count in the tables above from scratch (validation, survivorship, bounding, finer cohorts, imputation, geography, decomposition, habitual-core overlap, snapshot cross-validation, gender, representativeness index), independently of the analysis code; the ecological roll-off correlation is in `scripts/diag_wa_rolloff_2024.py` .

Appendix D — Related work

The finding — that low-salience electorates are older and smaller — is well established; this paper does not claim to discover it. Its contribution is narrower and twofold. First, it provides a **validated, individual-record** measurement of Washington's recent November electorates across the full salience gradient (presidential → midterm → off-year) from ~100% of the state's vote records, in a **universal vote-by-mail, same-day-registration** state where the usual access-cost explanation for youth drop-off is weaker. Second, and equally load-bearing, it confronts a common voter-file problem directly: because historical vote records are reconstructed from a *current* registration file, past voters who have since died, moved, or been canceled can be missing — an assumption many voter-file analyses leave implicit. By benchmarking against certified ballot counts, characterizing the observable attrition, and formally bounding the full residual, the paper shows the age-composition result is not an artifact of current-roll survivorship.

- **Turnout composition by salience (surge-and-decline).** Campbell, "Surge and Decline: A Study of Electoral Change," *Public Opinion Quarterly* 24(3) (1960): 397–418. Wolfinger & Rosenstone, *Who Votes?* (Yale, 1980); Leighley & Nagler, *Who Votes Now?* (Princeton, 2013) — age is among the most durable turnout predictors.
- **Off-cycle election timing, composition, and representation.** Anzia, *Timing and Turnout: How Off-Cycle Elections Favor*

Organized Groups (Univ. of Chicago Press, 2014); Hajnal & Trounstine, "Where Turnout Matters," *Journal of Politics* 67(2) (2005): 515–535; Hajnal, Kogan & Markarian, "Who Votes: City Election Timing and Voter Composition," *American Political Science Review* 116(1) (2022): 374–383; Kogan, Lavertu & Peskowitz, "Election Timing, Electorate Composition, and Policy Outcomes: Evidence from School Districts," *American Journal of Political Science* 62(3) (2018): 637–651; Lucero, Robles, Trounstine & Collins, "What Date Works Best for You? Changes in Electorate Demographics and Policy Priorities in Concurrent Elections," *Urban Affairs Review* (2025); Einstein, Palmer, Hamilton & Singer, "Age and Homeownership Drive the Local Turnout Gap," *Urban Affairs Review* (2025) — the closest analog to the age result here.

- **Contested policy effects of timing.** Ornstein, "Election Timing Revisited: Evidence from California's Voter Participation Rights Act" (working paper, 2024) — turnout/diversity gains but null downstream policy effects.
- **Down-ballot roll-off.** Wattenberg, McAllister & Salvanto, "How Voting Is Like Taking an SAT Test: An Analysis of American Voter Rolloff," *American Politics Quarterly* 28(2) (2000): 234–250 — roll-off is substantially information-driven.
- **Voter-file / individual-level method.** Ansolabehere & Hersh, "Validation: What Big Data Reveal About Survey Misreporting and the Real Electorate," *Political Analysis* 20(4) (2012): 437–459; Hersh, *Hacking the Electorate* (Cambridge, 2015).
- **Decomposition method.** Kitagawa, "Components of a Difference Between Two Rates," *JASA* 50(272) (1955): 1168–1194; Das Gupta, *Standardization and Decomposition of Rates* (U.S. Census Bureau, P23-186, 1993).

Appendix E — 65+ share of the electorate by county

The off-cycle senior tilt is not a King County artifact or a rural artifact: the presidential→off-year shift toward seniors is **positive in all 39 counties** (`scripts/verify_who_decides_wa.py` #24). Counties are sorted by their 2024 presidential 65+ share; the last column is the average off-year (2023, 2025) share minus the presidential share. The off-year average uses 2023 and 2025 because those years have substantially higher analyzable coverage than 2021 (99.6% and 95.9% vs 90.8%); including 2021 does not change the conclusion — averaged over all three off-years the gap remains positive in every county (King +7.5 to Franklin +16.1).

County	2024 Pres	2023 Off	2025 Off	Pres→Off	County	2024 Pres	2023 Off	2025 Off	Pres→Off
Jefferson	53.2%	66.1%	65.7%	+12.7	Douglas	34.3%	50.8%	53.2%	+17.7
Pacific	48.2%	60.1%	62.3%	+13.0	Cowlitz	32.5%	46.9%	49.7%	+15.8
Clallam	47.3%	59.9%	60.9%	+13.1	Kitsap	32.2%	44.1%	44.5%	+12.1
San Juan	47.3%	59.5%	57.8%	+11.4	Grant	32.0%	47.1%	48.3%	+15.7
Ferry	45.4%	55.7%	62.1%	+13.5	Yakima	31.5%	47.4%	48.1%	+16.3
Wahkiakum	45.0%	54.0%	56.9%	+10.5	Klickitat	31.3%	45.0%	44.5%	+13.4
Island	43.0%	54.5%	60.1%	+14.3	Thurston	30.9%	43.4%	42.8%	+12.2
Columbia	42.4%	51.8%	55.9%	+11.5	Adams	30.2%	43.2%	43.5%	+13.1
Garfield	42.4%	54.4%	50.5%	+10.1	Whatcom	29.6%	37.5%	40.0%	+9.1
Asotin	40.9%	57.8%	58.3%	+17.1	Spokane	29.6%	40.3%	43.2%	+12.2
Okanogan	40.9%	50.2%	55.1%	+11.8	Benton	29.1%	42.4%	44.9%	+14.6
Mason	40.0%	55.7%	55.7%	+15.7	Clark	27.9%	44.2%	42.3%	+15.4
Pend Oreille	39.9%	54.8%	58.6%	+16.8	Pierce	26.7%	39.9%	40.4%	+13.4
Lincoln	39.6%	51.4%	54.2%	+13.2	Whitman	26.3%	38.5%	40.4%	+13.1
Grays Harbor	39.4%	54.7%	57.6%	+16.8	Snohomish	25.5%	36.8%	38.4%	+12.1
Kittitas	38.6%	49.8%	54.1%	+13.3	Franklin	24.0%	39.8%	42.6%	+17.2
Stevens	38.4%	50.3%	56.0%	+14.7	King	23.0%	32.2%	30.7%	+8.4
Skagit	38.3%	52.3%	54.3%	+15.0					
Walla Walla	35.6%	48.6%	52.3%	+14.8					

County	2024 Pres	2023 Off	2025 Off	Pres→Off	County	2024 Pres	2023 Off	2025 Off	Pres→Off
Chelan	35.2%	45.2%	51.6%	+13.2					
Lewis	34.7%	50.0%	51.9%	+16.2					
Skamania	34.5%	48.9%	49.8%	+14.9					

The off-year 65+ share ranges from **30.7% (King)** to **66% (Jefferson)**, and every county moves toward seniors off-cycle (gaps +8.4 to +17.7 points). The result is a statewide phenomenon; King is simply its youngest instance.

Appendix F — Contest-level roll-off on the even-year ballot

The one thing this paper does *not* measure is whether a voter marked a specific contest. The obvious sequel — and the live objection to consolidation — is **down-ballot roll-off**: if local nonpartisan races moved onto the long even-year ballot, how many returned ballots would skip them? It is estimated from certified 2024 precinct returns (`scripts/diag_wa_rolloff_2024.py`), as (ballots counted − votes cast in a contest) / ballots counted (official 2024 ballots = 3,961,569).

Contest type (2024 general)	Example	Roll-off
Top of ticket	President	1.1%
Partisan statewide office	Governor … Insurance Comm.	2.7–6.8%
Statewide ballot measure	I-2109 … I-2124	4.1–5.6%
Nonpartisan statewide, contested	Supreme Court Pos. 2; Supt. of Public Instruction	**16.6–17.2%**
Nonpartisan statewide, uncontested	Supreme Court Pos. 8, 9	**33.7–34.4%**

Court of Appeals (regional — only one division votes) and Lt. Governor (loaded in only 5,355 of 8,111 precincts / 38 of 39 counties — a partial-load artifact, not roll-off) are excluded; see the script header.

Two honest points follow. First, Washington's even-year nonpartisan roll-off is not trivial: about **17%** in contested statewide nonpartisan contests and about **34%** in uncontested ones in 2024. These are plausible upper-bound analogs for local nonpartisan races, not direct estimates of city-council or school-board roll-off; voters generally have less information about judicial and local nonpartisan contests than about partisan statewide offices, and the age profile of Washington local-contest roll-off remains untested.

Second, even under large roll-off assumptions consolidation would probably enlarge the contest-level electorate — but the *size* of that enlargement depends on whether local races move onto a **presidential-** or a **midterm-year** ballot. Using Washington's recent official even-year turnout as the baseline (share of registered casting a vote in the local contest):

Scenario (baseline turnout)	5% roll-off	17% roll-off	34% roll-off
Presidential-year, 2024 (~79%)	~75%	~66%	~52%
Midterm-year, 2022 (~64%)	~61%	~53%	~42%
Odd-year, 2021/23/25 avg (~38%) — current baseline	—	—	—

The enlargement claim is strongest for presidential-year consolidation, still plausible for midterm-year consolidation, and weakest for low-information or uncontested local races placed on a *midterm* ballot (~42% vs the ~38% off-year baseline). The grid also treats roll-off as *fixed* across scenarios, which it is not: a presidential electorate contains more low-information peripheral voters — the ones who roll off most — so its true roll-off is likely higher than a midterm electorate's, making the presidential row an upper bound and the presidential–midterm gap narrower than shown. The better conclusion is not that roll-off is second-order in every case, but that even substantial roll-off does not erase the turnout advantage unless local-contest roll-off runs far higher than the contested-nonpartisan benchmark.

Does roll-off skew young? An ecological first look. The concern behind this objection is that the young voters consolidation would add are also the ones most likely to skip a local contest. Cast-vote records cannot answer this directly — ballots are anonymous and carry no voter age, so the roll-off *age profile* is unmeasurable at the individual level under secret-ballot rules, and an ecological cut is the ceiling. Across the 39 counties, roll-off on that contest is if anything *higher* where the electorate is older (Pearson r ≈ +0.6, **uncorrected for urbanicity**) — that is, the county pattern does not show younger places skipping the contest more. This is weak evidence: it is ecological, confounded with the rural/urban gradient (older counties are rural, with thinner coverage of statewide judicial races), and measured on a *statewide* judicial race that is an imperfect analog for hyperlocal contests. It points away from young-concentrated roll-off but cannot establish individual behavior. A finer precinct-level ecological analysis is the natural sequel, and the next paragraph runs it; an individual-level administrative test is not available from public cast-vote records, because ballot secrecy prevents linking contest choices to voter age.

The precinct-level cut, net of urbanicity. The county correlation's weakness is that it cannot separate age from the rural/urban gradient — older counties are rural — so the +0.6 conflates the two, as the county cut itself flagged. Running the same measure across the ~4,900 precincts with both a 2024 presidential count and apportioned ACS demographics (`scripts/diag_wa_rolloff_precinct.py`) separates them. Precinct roll-off on Supreme Court Pos. 2 — here measured against the presidential vote rather than total ballots — averages 16.4% (16.5% ballot-weighted), close to the 16.3% statewide same-denominator figure (the 17.2% in the table above is against total ballots). The raw sign holds but the strength does not. Measured on the precinct's own 2024 electorate — the *same* predictor as the county's +0.6 — the association falls to Pearson r ≈ +0.09, under a sixth of the county value, once finer geography strips out the aggregation that inflated it. On the resident population it is r ≈ +0.26 on the ACS 65+ share (+0.28 on median age), still modest and still positive. The payoff of the finer geography is that these units can carry controls: residualizing roll-off and the ACS 65+ share on precinct income, education, median home value, renter share, and (log) population — a rough proxy for the urban/rural and SES gradient — cuts the association to a partial r ≈ +0.11 for the contested court race and +0.02 for Superintendent of Public Instruction. The strong county figure thus survives *neither* disaggregation *nor* control; what remains is small and, where nonzero, still runs *against* the objection — older places skip the contest slightly more, not less. The result is firmer than the county version but unchanged in direction: the data give no support to the fear that the young voters consolidation would add are the ones who would skip the local contest. The caveats compound rather than lift — the cut is still ecological and cannot speak to individuals; the urbanicity proxy is imperfect (precinct populations are drawn to be roughly equal, so a precinct-area density measure from the geometry files would sharpen it); and a statewide judicial race is a loose analog for hyperlocal contests — but every refinement has pushed the same way. (Because it apportions ACS block groups to precincts and uses the VRDB precinct crosswalk, this cut relies on build inputs beyond the public certified returns; see the script header.)

End note — data, reproduction, and series

Data. Washington's statewide voter-registration database (April 2026 extract): the 5.51M-voter roll joined to 27.1M individual vote records and each voter's year of birth (`data/wa_vrdb.duckdb`); access terms in Appendix B. Official ballot counts are the certified statewide totals published by the WA Secretary of State (`results.vote.wa.gov`). Adult-resident age composition is the U.S. Census American Community Survey 2020–2024 5-year, table B01001 (Washington, FIPS 53).

Institutional context. Washington is an unusually informative case because formal ballot-access costs are lower than in many states: registered voters receive mail ballots, which can be returned by mail without postage or by drop box; eligible voters may register or update registration in person through 8 p.m. on Election Day; and registration is automatic through qualifying agency transactions (WA Secretary of State). Those rules do not eliminate every participation barrier — information costs, mobility, local attachment, address stability, and unequal political recruitment still matter — but they make it harder to attribute the age gap primarily to ballot-access friction.

Reproduction. `scripts/verify_who_decides_wa.py` re-derives every count in the tables above from scratch (sections #1–#29, incl. the county breakdown, habitual-core overlap, snapshot cross-validation, gender, and representativeness index); the roll-off appendix and its ecological age correlation are in `scripts/diag_wa_rolloff_2024.py` , with the finer precinct-level, urbanicity-controlled cut in `scripts/diag_wa_rolloff_precinct.py` ; `scripts/diag_wa_individual_findings.py` and `scripts/diag_turnout_decomposition.py` produce the underlying figures; and `scripts/acs_wa_adult_age.py` reproduces the adult-resident and CVAP rows from the Census API.

Series. Lead paper of the electoral-health series (with `electoral-health-whitepaper.md` and `cross-state-fec-money.md`). Party-resolved companions in states that publish party of record: `who-decides-new-york.md` (deep blue) and `who-decides-idaho.md` (deep red).

Companion paper: Safe-Seat Washington — once in an off-year general, how often is the contest even a choice? It counts **observed** competitiveness of every partisan legislative + congressional seat, 2016–2024 (≈85% non-competitive), and extends the count to a four-state lower-chamber map.